Logic in Computer Science

Problem Sheet 5: Logical Consequence

September 20, 2025

1. Check whether v satisfies every element in U for a given v and U.
   (a) $U = \{(p_3 \to p_1), (p_1 \to (p_2 \vee p_1)), (p_3 \wedge (p_1 \vee p_2)), (p_2 \leftrightarrow p_1)\}$ and $v(p_1) = T, v(p_2) = T, v(p_3) = F$
   (b) $U = \{p_1, (p_1 \to p_2), (p_2 \to (\neg p_)), (\neg(p_3 \leftrightarrow p_2)\}$ and $v(p_1) = T, v(p_2) = T, v(p_3) = T$
2. Check whether the following set of formulas are satisfiable.
   (a) $U_1 = \{(p_1 \to p_2), (p_1 \to (p_2 \to p_1)), (\neg(p_2 \to p_1))\}$
   (b) $U_2 = \{(p_1 \vee p_2), ((\neg p_2) \vee (\neg p_3)), (p_3 \vee p_4), ((\neg p_4) \vee (\neg p_5)), \dots\}$
3. Show that $U \models A$ for a given U and A.
   (a) $U = \{(p \to (q \to r)), (p \to q), p\}$ and $A = r$
   (b) $U = \{(p \to q), (q \to r)\}$ and $A = (p \to r)$
   (c) $U = \{p, (p \to q), (q \to (\neg p)), (\neg(r \leftrightarrow q))\}$ and $A = (r \vee q)$
4. *Prove* or *disprove* the following statements. Here U is a set of well formed formulas.
   (a) $\{A\} \models B$ iff $A \equiv B$
   (b) $U \not\models A$ then $U \models \neg A$
   (c) $U \models A$ then $U \not\models \neg A$
   (d) $U \models A$, then $U \setminus \{B\} \models A$.
   (e) If $U \models A$, then $Mod(U) \subseteq Mod(\{A\})$
5. *Prove* or *disprove* the following statements. Here U_1, U_2 are sets of well formed formulas.
   (a) If $U_1 \subseteq U_2$, then $Mod(U_2) \subseteq Mod(U_1)$
   (b) If $U_1 \subseteq U_2$, then $Mod(U_1) \subseteq Mod(U_2)$
   (c) If $U_1 \models A$ and $U_1 \subseteq U_2$, then $U_2 \models A$
6. Prove the following statements.
   (a) $U \models A$ iff $U \cup \{\neg A\}$ is unsatisfiable.
   (b) $U \models (A \to B)$ iff $U \cup \{A\} \models B$.
   (c) $U \models A$ and B is a valid formula, then $U \setminus \{B\} \models A$.